

You completed this test on *30/03/2022, 09:32*
Your score is 92.31%

CORRECT

How would you create the identity matrix in python?

✓ ***np.eye(3)*** Syntax

identity(3)

np.array([1,0,0], [0,1,0], [0,0,1])

All of the above

CORRECT

Suppose you want to join train and test dataset (both are two numpy arrays train_set and test_set) into a resulting array (resulting_set) to do data processing on it simultaneously. How would you join the two arrays to get the resulting set?

```
train_set = np.array([1, 2, 3])
test_set = np.array([[0, 1, 2], [1, 2, 3]])
resulting_set --> [[1, 2, 3], [0, 1, 2], [1, 2, 3]]
```

resulting_set = train_set.append(test_set) We will use vertical stack or vstack to join dataset

resulting_set = np.concatenate([train_set, test_set])

✓ ***resulting_set = np.vstack([train_set, test_set])***

None of these

CORRECT

Imagine, you have a dataframe train file with 2 columns & 3 rows, which is loaded in pandas.

train = pd.DataFrame({'id':[1,2,4],'features':[["A","B","C"],["A","D","E"],["C","D","F"]]})

Now you want to apply a lambda function on “features” column:

train['features_t'] = train["features"].apply(lambda x: " ".join(["_".join(i.split(" ")) for i in x]))

What will be the output of the following print command (Assume \n is newline) ?

print(train['features_t'])

- ✓ ***A B C \n A D E \n C D F***
- A B \n A D E \n C D F
- Error
- None of these

CORRECT

Consider a function “fun” which is defined below:

```
def fun(x):
x[0] = 5
return x
```

Now you define a list which has three numbers in it.
g = [10,11,12]

Which of the following will be the output of the given print statement: print fun(g), g ?

It changes the original data too

- ✓ ***[5, 11, 12] [5, 11, 12]***
- 5, 11, 12] [10, 11, 12]
- [10, 11, 12] [10, 11, 12]
- [10, 11, 12] [5, 11, 12]

CORRECT

Suppose you are given a data frame df.

df = pd.DataFrame({'Click_Id':['A','B','C','D','E'],'Count':[100,200,300,400,250]})

Now, you want to assign a df to df1, so that you can recover original content of df in future using df1 as below.

df1 = df

Now you want to change some values of “Count” column in df.

df.loc[df.Click_Id == 'A', 'Count'] += 100

Which of the following will be the right output for the below print statement?

print(df.Count.values,df1.Count.values)

It is same as prev one both df and df1 points to same address location

✓ ***[200 200 300 400 250] [200 200 300 400 250]***

[100 200 300 400 250] [100 200 300 400 250]

[200 200 300 400 250] [100 200 300 400 250]

None of these

CORRECT

Suppose, you are given 2 list – City_A and City_B.

City_A = [‘1’,’2’,’3’,’4’]

City_B = [‘2’,’3’,’4’,’5’]

In both cities, some values are common. Which of the following code will find the name of all cities which are present in “City_A” but not in “City_B”.

✓ ***[i for i in City_A if i not in City_B]***

it should only print the contents of CityA

[i for i in City_B if i not in City_A]

[i for i in City_A if i in City_B]

None of these

CORRECT

Suppose you are trying to read a file “temp.csv” using pandas and you get the following error.

Traceback (most recent call last):

File "<input>", line 1, in<module>

UnicodeEncodeError: 'ascii' codec can't encode character.

Which of the following would likely correct this error?

pd.read_csv(“temp.csv”, compression=’gzip’)

utf-8 encoding should be used as ascii code doesnt recognize some characters

pd.read_csv(“temp.csv”, dialect=’str’)

✓ ***pd.read_csv(“temp.csv”, encoding=’utf-8’)***

None of these

CORRECT

Suppose you are defining a tuple given below:

tup = (1, 2, 3, 4, 5)

Now, you want to update the value of this tuple at 2nd index to 10. Which of the following option will you choose?

tup(2) = 10

Tuples are immutable cannot change values

tup[2] = 10

tup{2} = 10

✓ ***None of these***

CORRECT

Which of the following is a correct implementation of the mean squared error (MSE) metric?

return np.mean((np.square(real_target) – np.square(predicted_target)))

(y-ypred)^2

✓ ***return np.mean((real_target – predicted_target)**2)***

return np.sqrt(np.mean((np.square(real_target) – np.square(predicted_target)))

None of the above

CORRECT

You want to read a website which has url as “www.abcd.org”. Which of the following options will perform this task?

urllib2.urlopen(www.abcd.org)

in python both packages can read a website

requests.get(www.abcd.org)

✓ ***Both A and B***

None of these

INCORRECT

List out the critical assumptions of linear regression

The relationship between X and Y is Linear (Linearity)
The variance of residual is the same for any value X (Homoscedasticity)
Observations are independent to each other (Independence)
For any value of X, Y is normally distributed (Normality)

CORRECT

Is Logistic regression mainly used for Regression?

Yes

✓ ***No***

Logistic regression is used for classification

CORRECT

Is Linear Regression Parametric/Non Parametric approach?

✓ ***Parametric***

LR requires parameter

Non-Parametric

CORRECT

Is it possible to apply a logistic regression algorithm on a 3-class Classification problem?

✓ ***Yes***

we can classify many classes too

No

CORRECT

Calculate 1) Accuracy 2) Precision 3) Recall 4) F1 score for the below data:

	Actual Class : 1	Actual Class : 0
Predicted Class : 1	85	890
Predicted Class : 0	15	10

Accuracy = [✓ ***9.5%***]
Precision = [✓ ***0.087***]
Recall = [✓ ***0.85***]
F1 Score = [✓ ***0.16***]

using the formula
accuracy = (TP+TN)/(TP+TN+FP+FN)
precision = TP/(TP+FP)
recall = TP/(TP+FN)
f1 = 2*P*R / (P+R)

CORRECT

Which of the following methods do we use to best fit the data in Logistic Regression?

Least Square Error

✓ ***Maximum Likelihood***

Jaccard distance

Both A and B

CORRECT

In regression analysis, which of the statements is true?

✓ ***The mean of residuals is always equal to Zero.***

The Mean of residuals is less than Zero at all times.

The Mean of residuals is more than Zero at all times.

You do not have any such rule for residuals.

CORRECT

Suppose a classifier classifies 27000 examples correctly and 3000 examples incorrectly. Compute the error by hand.

3000/30000 = 0.1

✓ ***0.1***

0.111

0.9

0.5

CORRECT

Accuracy and error measured on the same dataset always sum to 1

✓ ***True*** acc + loss/error = 1

False

CORRECT

Which of the following is an example of overfitting? as training error is less and validation error is more the model is overfitting

✓ ***Classification error of 0.2 in training set and classification error of 0.6 on validation set***

Classification error of 0.8 in training set and classification error of 0.6 on validation set

Classification error of 0.2 in training set and classification error of 0.2 on validation set

Classification error of 0.5 in training set and classification error of 0.4 on validation set

INCORRECT

Which of the following can be used to prevent overfitting? Select all that apply

- ✓ ***Cross-validation***
- ✓ ***Remove few features***

All of the given option

- **Regularization**
- **Add Noise to data to include linearity**
- **Add Noise to data to include non-linearity**

CORRECT

Once shuffling and splitting of the dataset is done, which of the following is not correct in k-Fold cross-validation?

- Take the group as a hold out or test data set
- Take the remaining groups as a training data set
- Fit a model on the training set and evaluate it on the test set
- ✓ ***Retain the evaluation score and keep the model***

We wont retain the score

CORRECT

According to bias-variance tradeoff Increasing the bias will decrease the variance

bias n variance are inversely proportional

- ✓ ***True***
- False

CORRECT

What will be the train accuracy of the model in case of overfitting?

- Low
- ✓ ***High***

train high validation low

CORRECT

What is the name given to the process of splitting the train data into train set and validation set?

- ✓ ***Cross Validation***

Regularization

Clustering

Regression

CORRECT

What does regularization do?

✓ ***Reduces variance of the model without substantial increase in its bias***

Increases variance of the model without substantial increase in its bias

Reduces variance of the model with substantial increase in its bias

Increases variance of the model with substantial increase in its bias

Passed. Let's continue.